A GEOSPATIAL DECISION-SUPPORT FRAMEWORK FOR URBAN AND SUBURBAN PHOTOVOLTAIC DEPLOYMENT

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ABSTRACT:
The transition towards renewable energy in Italy is progressing at a slower pace than required, partly due to a structural lack of coherent, up-to-date, and easily accessible geospatial information to support investment decision-making.
This research is conducted within the *Space It Up* project, funded by the Italian Space Agency (ASI) and the Ministry of University and Research (MUR), and aims to develop an operational tool to support industrial stakeholders in the identification of suitable sites for photovoltaic deployment by integrating Earth Observation (EO) data, geospatial information, regulatory constraints, incentives, and system productivity.
The methodology is applied to a case study in Regione Lombardia, in the context of the "Aree Idonee" decree issued in July 2024, which defines criteria for the identification of suitable and unsuitable areas for renewable energy installations and introduces simplified authorization procedures for photovoltaic plants in specific zones. Accordingly, the regulatory framework established by the decree is explicitly incorporated into the analysis. Official datasets from Regione Lombardia, the Ministry of the Environment and Energy Security (MASE), CORINE Land Cover, Copernicus products, and other EO-derived data are jointly exploited.
In addition, the study addresses rooftop characterization through the development of a multispectral approach for detecting existing photovoltaic installations, enabling the assessment of retrofit potential, the identification of suitable surfaces for solar energy production, and the evaluation of industrial areas for rooftop photovoltaic deployment.

1. INTRODUCTION

This project is framed within the European policy context defined by the commitment to reduce greenhouse gas emissions by 55% by 2030 and to achieve climate neutrality by 2050 under the *Fit for 55* package. In parallel, in response to recent energy challenges the European Union launched the *REPowerEU* programme to strengthen energy security and reduce external dependencies. This policy sets a target of 45% renewables by 2030 and promotes simplified authorization procedures for new solar installations.
The Integrated National Energy and Climate Plan (PNIEC, 2024) identifies photovoltaics as a key technology for achieving European targets and explicitly calls for the development of tools and methodologies for mapping photovoltaic potential, including solar cadastres, at both urban and national scales.
An assessment of existing tools reveals the widespread availability of solar cadastres, particularly in urban contexts, including initiatives implemented in Switzerland, Tirol, Wien, St. Pölten, Podgorzyn and Chrastava, London, Calgary, Bolzano, Baden, Hannover, Hessen, Amsterdam, and other European and non-European cities. These instruments typically provide building-scale assessments of rooftop solar potential.
Given the relevance and timeliness of the topic, this study aims to develop an operational mapping tool that integrates solar potential with the set of regulatory, spatial, and technical information. To this end, key stakeholders of the photovoltaic sector operating in Italy were consulted to identify the main criteria guiding site scouting activities. The interviews indicate that the primary assets of interest are ground-mounted photovoltaic systems exceeding 10 hectares and rooftop installations with a surface area larger than 1,000 m².
This work focuses on two main aspects:
– *Permitting*: the collection and spatial representation of regulatory incentives and constraints within a georeferenced framework (Section 2),
– *Detection of existing photovoltaic systems*: the development of a method to identify rooftop-mounted solar installations using Sentinel imagery (Section 3).

2. PERMITTING

Authorization procedures for photovoltaic systems are governed by the "Aree Idonee" Decree, which defines four distinct authorisation regimes (Table 1). Under this framework, areas classified as suitable are subject to simplified and accelerated authorisation procedures, whereas areas identified as unsuitable, are subject to more restrictive regulatory conditions.

Table 1. Authorisation regimes defined by the "Aree Idonee" Decree.

Class	Authorisation regime	Regulation reference
a)	Suitable Area	Legislative Decree No 190, 25 November 2024, N. 190, Art 11-bis
b)	Unsuitable Area	«Linee guida per l'autorizzazione degli impianti alimentati da fonti rinnovabili» (2010)
c)	Ordinary Area	Not specified
d)	Forbidden Area	Legislative Decree No 190, 25 November 2024, N. 190, Art 11-bis

2.1 Data and Methods

Data were retrieved from the Geoportale Lombardia, the "Carta Nazionale di Consumo del Suolo 2023" (CNCS2023) compiled by ISPRA, and the CORINE Land Cover 2018 (CLC2018) dataset, and subsequently processed using QGIS software. The

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most recent source available for each dataset was selected. The temporal discrepancies are considered acceptable for the purposes of the preliminary site-screening process. The prototype was developed for the Regione Lombardia case study area.

2.1.1 Suitable areas

Areas classified as suitable pursuant to Article 20, paragraph 8 of Legislative Decree No. 199/2021 include:

a) *Existing photovoltaic installations undergoing modification or refurbishment:*
Corresponding to "ground mounted photovoltaic installations" class of CNCS2023 dataset. The detection of rooftop-mounted systems is addressed separately in Section 3.

b) *Sites subject to environmental remediation as defined under Title V, Part IV of Legislative Decree No. 152 of 3 April 2006:*
Obtained from AGISCO database (Registry and Integrated Management of Contaminated Sites).

c) *Quarries and mines that are no longer in operation, have not been reclaimed, have been abandoned, or are in a state of environmental degradation:*
Closed quarries with an area larger then 10.000 m2 were selected from the quarry registry provided by Geoportale Lombardia.
A distinction was made between dry and water-filled quarries, as the latter may be suitable for floating photovoltaic installations:
– Closed dry quarries: Quarries intersecting areas classified as "non-renaturalized extraction areas" in the CNCS2023 dataset and as "mineral extraction sites" in the CLC2018 dataset were selected. This category includes quarries with more than 50% of their surface not flooded or with a non-water surface exceeding 15,000 m².
– Closed submerged quarries: Areas classified as water bodies in the DUSAF 6.0 (2018) dataset were extracted from the regional quarry registry. Only quarries with a submerged area larger than 20,000 m² were considered.

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2) Areas within industrial plants and agricultural areas enclosed within a perimeter whose boundary points are located no more than 500 m from the same plant:
Corresponding to industrial buildings and their surrounding areas within a 500 m buffer, intersecting land classified as agricultural in the CLC2018 dataset.
3) Areas adjacent to the motorway network within a maximum distance of 300 m.
A 300 m buffer around motorway was generated to delineate agricultural areas potentially suitable for photovoltaic deployment.

2.1.2 Unsuitable areas

Areas deemed unsuitable for the installation of photovoltaic systems are defined by Regional Council Resolution (Giunta Regionale) No. 9/3298 of 18 April 2012, Part V, in accordance with the Ministerial Decree of 10 September 2010. While photovoltaic installations are not explicitly prohibited in these areas, they are not eligible for accelerated or simplified authorisation procedures. The classification of unsuitable areas accounts for landscape, environmental, cultural, hydrogeological, and infrastructural constraints.

2.1.3 Integrated Suitability Analysis

Suitable and unsuitable areas are subsequently integrated into a single map, supporting the identification of locations most appropriate for photovoltaic deployment and allowing a suitability analysis. Figure 1 illustrates the results for a representative case study area, where several agricultural parcels fall within the buffer zones of highways or industrial facilities (shown in green) but are simultaneously included within a PLIS protected park (shown in orange). As a result, feasibility assessments should focus on agricultural areas that are not subject to park-related constraints.

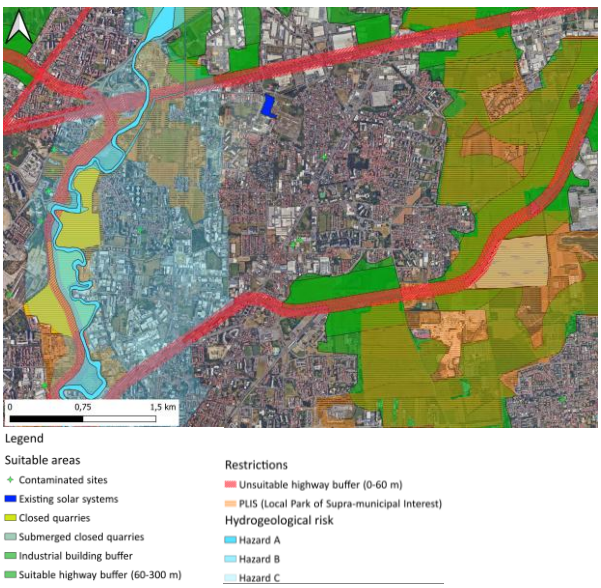


Figure 1. Spatial overlap between areas classified as suitable and unsuitable for photovoltaic installation.

3. DETECTION OF EXISTING SOLAR SYSTEMS ON ROOFTOP

Rooftop surfaces represent a significant opportunity for solar energy deployment while avoiding additional land take. Information on existing rooftop-mounted photovoltaic systems is analysed and integrated into the mapping framework. To this end, a method for detecting existing installations is developed to identify available roof surfaces. The presence of rooftop photovoltaic systems is also used as an indirect proxy for grid connectivity.

3.1 Data and methods

The datasets used for this analysis include a Sentinel-2B L2A image tile 32TNR acquired on 19 June 2025 and polygonal data for commercial and industrial buildings obtained from the Database Geo-Topografico (DBT) of Regione Lombardia, limited to features with a surface area greater than 10,000 m².
Spectral signatures were extracted for 411 buildings from the Sentinel-2B imagery using SNAP software. The sample includes both buildings equipped with rooftop photovoltaic systems and buildings without rooftop installations. Samples were selected to ensure a uniform spatial distribution across industrial areas and were manually labelled as "simple roof" or "solar panel" based on visual interpretation of Qgis basemaps Google satellite and Bing satellite, which have generally an update time of months, but in the worst cases it can go up to three years. Nevertheless, the comparison of the two basemaps was the best option available, as the most recent orthophoto dates to 2021.
An average spectral signature representative of photovoltaic panels was derived from the point samples extracted in SNAP (Figure 2). The resulting average signature is comparable to the

theoretical signature defined in the "Roof Material Spectral Library" (C. Wyard et al., 2023). Both are characterised by low reflectance in the visible and near-infrared regions, followed by a peak in the B11 band (SWIR1) and a decrease in the B12 band (SWIR2). Hyperspectral datasets additionally reveal an absorption feature at approximately 1730 nm, which cannot be resolved by the spectral bands of Sentinel-2.

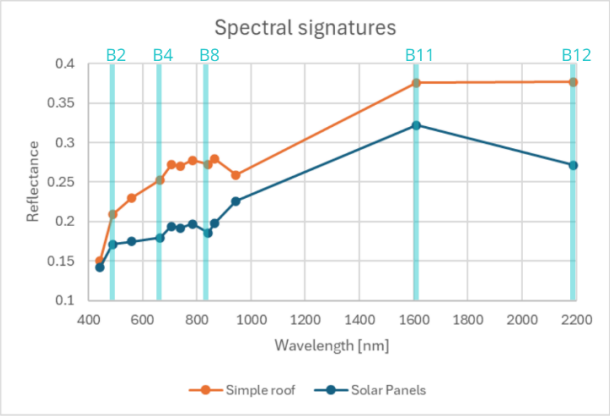


Figure 2. Average Sentinel-2 spectral signature of rooftop-mounted photovoltaic panels. Bands used in the indices are highlighted in light blue.

3.1.1 Classification model

The classification model was developed by combining three spectral indices selected for their compatibility with Sentinel-2 spectral bands and their operational capability to discriminate between rooftops with and without photovoltaic installations.

Enhanced Photovoltaic Index (EPVI)

The Enhanced Photovoltaic Index was introduced by Wang *et al.* (2024) to detect photovoltaic installations by emphasising the reflectance peak in the SWIR1 band (B11) relative to the near-infrared (B8) and SWIR2 (B12) bands:

$$EPVI = \frac{2 \cdot B11 - B12 - B8}{2 \cdot B11 + B12 + B8} \quad (1)$$

Solar Photovoltaic Index (SPI)

The Solar Photovoltaic Index was originally developed by Tian *et al.* (2022) for Landsat-8 OLI imagery and is here adapted to Sentinel-2 spectral bands. Like EPVI, this index exploits the reflectance peak observed in the B11 band to enhance the spectral response of photovoltaic panels:

$$SPI = \frac{(2 \cdot B11 - (B8 + B12)) - |\, B8 - B4 \,| - |\, B8 - B12 \,|}{2 \cdot B11 + (B8 + B12)} \quad (2)$$

B2/B4 ratio

The B2/B4 ratio exploits the relatively higher reflectance of photovoltaic panels in the blue band (B2) compared to the red band (B4):

$$B2/B4 = \frac{B2}{B4} \quad (3)$$

The three indices are based on bands with geometric resolution of 10 m or 20 m, resampled at 10 m. For each index, the discriminative capability was evaluated based on the separation between classes, quantified as the difference between the 75th percentile of the "simple roof" class and the 25th percentile of the "rooftop photovoltaic" class. Since all indices exhibit higher values for photovoltaic panels, this metric was used to assess their suitability for rooftop PV detection within the proposed framework. The three indices are used to classify the pixels in Solar Panel (PV) and Simple rooftop (T), on the base of the distance from centroids of solar panel sample and rooftop sample. Some pixels presenting comparable distance are considered Non-classified (NC).

3.1.2 Results

Validation was performed using an independent a posteriori sample, yielding an overall accuracy of 0.83 and a Cohen's Kappa coefficient of 0.66. User accuracy and Producer accuracy are reported in Table 2 and Table 3. The metrics are computed from around 160 samples per class, disregarding the non-classified pixels.

Table 2: User accuracy of validation sample

		Real classes	
		Solar panel	Simple roof
Classification	Solar Panel	74%	26%
	Simple roof	7%	93%

Table 3: Producer accuracy of validation sample

		Real classes	
		Solar Panel	Simple roof
Classification	Solar Panel	91%	22%
	Simple roof	9%	78%

Figure 3 shows an example of the resulting classification at the pixel level, representing PV pixels in blue, T pixels in orange and NC pixels with black.

An index representing the expected presence of rooftop photovoltaic installations (PAF) was computed for each building to aggregate pixel-level classification results at the building scale. To this end, the proportion of pixels assigned to each class was calculated. The index formulation accounts for the performance of the classification model by weighting each class according to its user accuracy. The PAF index was computed as:

$$PAF = PV_perc \cdot 0.74 + NC_perc \cdot 0.28 + T_perc \cdot 0.07 \quad (4)$$

Figure 4 illustrates the spatial distribution of the PAF values, with higher values represented by darker blue tones and lower values by lighter shades.

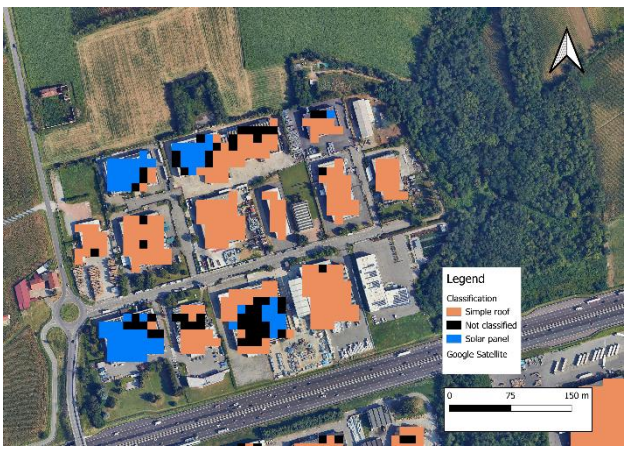


Figure 3. Building-level classification of the 10 m pixels

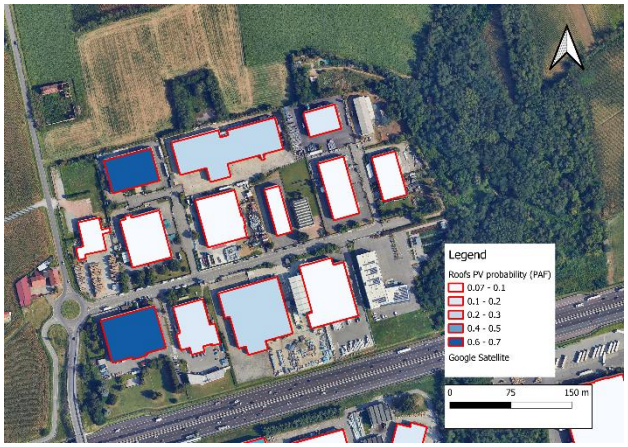


Figure 4. Building-level distribution of the PAF index derived from rooftop photovoltaic detection.

3.1.3 Robustness evaluation

The same classification approach was applied to the 8 buffer tiles of tile 32TNR, to cover the whole Lombardy region. The selected scenes date to summer 2025, as well. To assess the classification accuracy, a validation sampling was conducted. Around 50 pixels per tile were sampled among rooftops in Lombardy area, with user accuracy reported in Table 4. Only UA of Solar Panels class is assessed, as it is the class with lowest accuracy. The UA is quite variable among the tiles, but the weighted average of 77% is very close to the UA of tile 32TNR, confirming the robustness of the method for Sentinel-2.

Table 4: User accuracy of solar panel class performed with buffer tiles. Tiles are indicated with the last two letters, i.e. 32TPQ is PQ.

Tile	**PR**	**PQ**	**PS**	**NS**	**MS**
UA Solar panel %	71	72	88	84	60
Sampling points	55	47	50	49	50
Tile	**MR**	**MQ**	**NQ**	**All tiles**	
UA Solar panel %	67	86	90	77	
Sampling points	48	50	50	399	

4. CONCLUSIONS

This study demonstrates the integration of heterogeneous geospatial datasets into an operational framework supporting industrial-scale energy transition. The work successfully gathered all the indications required by current regulations in a single tool, and, therefore, it facilitates a potential user in efficiently identifying suitable sites, accelerating the scouting process. It is remarked that the selected datasets refer to different sources and inevitably present different spatial resolutions and temporal updates, although only official institutional datasets were used, and the most up-to-date version available for each source was selected. The framework should be updated whenever new data are released or when relevant regulatory provisions are modified.

Furthermore, a methodology for detecting rooftop-mounted photovoltaic systems was developed to identify available roof surfaces while providing an indirect proxy for grid connectivity, which is often difficult to assess using publicly available data. The proposed approach yields results that are adequate for the intended operational context; however, its ability to discriminate between photovoltaic installations and roof materials with similar spectral characteristics remains limited. This constraint could be addressed through the integration of hyperspectral data, such as PRISMA imagery. Another clear limit is the coarse spatial resolution of S2 bands, compared to the relatively small size of the rooftops, which influences the classification with border effects. Future developments will also consider additional information on electrical grid connectivity and roof geometry derived from LiDAR data, to assess sites suitability for photovoltaic panels installation.

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